

Bike Store Sales Performance Analysis

Project Overview

This project focuses on analysing the sales performance of a bike store business operating across three different stores located in different states. The analysis was performed using Excel and Power BI to identify trends, compare store performance, and understand possible reasons behind the underperformance of one specific store.

The project mainly investigates the performance of Rowlett Bikes, which consistently generated lower revenue, fewer orders, and fewer customers compared to the other two stores.

Tools & Technologies Used

- Power BI
 - Power Query
 - DAX
 - Excel
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Dataset Information

Dataset Used: Bike Store Relational Database

The dataset contains information related to:

- Orders
- Customers
- Products
- Categories
- Stores
- Staffs
- Stocks

The analysis was performed using data from 2016 to 2018.

Business Problem

The business operates across three stores:

- Baldwin Bikes
- Santa Cruz Bikes
- Rowlett Bikes

Among all three stores, Rowlett Bikes consistently underperformed in terms of:

- Revenue
- Orders
- Customers

The goal of this project was to:

1. Compare store performance across years.
 2. Identify trends and possible reasons behind lower revenue.
 3. Analyse customer and order distribution across states.
 4. Recommend business strategies for improving store performance.
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Key KPIs

The dashboard tracks the following KPIs:

- Total Revenue
 - Total Orders
 - Total Customers
 - Average Order Value (AOV)
 - Average Discount
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Dashboard Pages

1. Overview Dashboard

This page provides a high-level overview of overall business performance.

Insights:

- Revenue peaked in 2017 and dropped significantly in 2018.
 - Baldwin Bikes remained the highest-performing store across all years.
 - Rowlett Bikes consistently generated the lowest revenue.
 - Lower customer count and fewer orders might be one of the possible reasons behind Rowlett Bikes' lower performance.
 - The business generated 7.69M total revenue from 1,615 orders.
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2. Rowlett Bikes Performance Analysis

This page focuses specifically on the underperforming store.

Insights:

- Rowlett Bikes showed improvement in 2017 but experienced a major decline in 2018.
 - Several months in 2018 recorded zero transactions.
 - Lower customer count and lower order volume impacted yearly revenue.
 - Monthly sales trends showed inconsistent performance.
 - Average Order Value remained relatively stable, suggesting customer acquisition may be a larger challenge.
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3. Sales & Customer Insights

This page analyzes customer, order, product, and regional performance.

Insights:

- NY generated the highest customers, orders, and revenue.
 - TX recorded comparatively lower customer and order volume.
 - Lower market demand in TX might be one of the possible reasons behind Rowlett Bikes' weaker performance.
 - Road Bikes and Mountain Bikes were among the top-performing categories in TX.
 - Certain cities generated better revenue despite overall lower sales.
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4. Recommendations & Business Suggestions

This page provides business recommendations based on the analysis.

Recommendations:

1. Investigate 2018 Transaction Gap

- Improve transaction tracking systems.
- Monitor monthly sales activity regularly.
- Investigate possible operational or data-entry issues.

2. Increase Customer Engagement in TX

- Run local marketing campaigns.
- Improve customer acquisition strategies.
- Launch seasonal promotions and offers.

3. Promote High-Performing Categories

- Focus more on Road Bikes and Mountain Bikes.
- Maintain sufficient stock for popular products.
- Promote top-performing products in TX.

4. Focus on Higher-Revenue Cities

- Prioritize marketing in stronger-performing cities.
 - Improve regional targeting strategies.
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Data Cleaning & Preparation

The following steps were performed before analysis:

- Checked for missing values
 - Verified relationships between tables
 - Created data model in Power BI
 - Created calculated measures using DAX
 - Formatted and cleaned columns for reporting
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Skills Demonstrated

- Data Cleaning
- Data Modeling
- SQL Querying

- Data Visualization
 - KPI Analysis
 - Business Insights Generation
 - Dashboard Design
 - Storytelling with Data
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Conclusion

The analysis identified Rowlett Bikes as the weakest-performing store across all years. The store consistently recorded lower customers, fewer orders, and lower revenue compared to the other stores.

The project also highlighted a significant transaction gap during several months in 2018, which may indicate operational or reporting issues.

The dashboard provides data-driven recommendations focused on customer acquisition, product promotion, and regional targeting to improve future business performance.

GitHub Project Highlights

- Interactive Power BI dashboard
- Multi-page dashboard design
- KPI tracking and trend analysis
- Business recommendations based on insights